Bachelor of Computer Applications

Semester – V

Business Analytics

Assignment No.: 05

Title: Building Customer Analysis Dashboard Using Power BI

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Building Customer Analysis Dashboard Using Power BI

Problem Statement

A retail business stores customer information and order transactions in separate CSV files. Management requires a centralized dashboard to understand product-category popularity, payment preferences, revenue trends, and customer purchasing behaviour. The task is to import and connect the customer and order datasets in Power BI, create suitable measures, and present the analysis using appropriate visualizations.

Objectives

- Import the customer and order datasets into Power BI Desktop.
- Establish a one-to-many relationship between Customers and Orders using CustomerID.
- Create measures for total revenue, total orders, average order value, total quantity, and purchasing customers.
- Develop KPI cards and charts to answer the five questions specified in the assignment.
- Present the results in a clear and consistent dashboard.

Datasets Used

Two CSV datasets were used for this assignment:

- customer_analysis_customers_1000.csv — contains CustomerID, CustomerName, Gender, Age, City, State, IncomeBand, SignupDate, and CustomerSegment.
- customer_analysis_orders_5000.csv — contains CustomerID, OrderID, OrderDate, PaymentMethod, Product, ProductCategory, Quantity, DiscountPercent, and SalesAmount.

CustomerID is the common field used to connect customer details with order transactions.

Methodology / Steps Followed

Step 1: Data Import

- Opened Power BI Desktop and used Home → Get Data → Text/CSV to import both datasets.
- Loaded the customer and order tables into the Power BI data model.

Step 2: Data Modeling

- Connected Customers[CustomerID] to Orders[CustomerID] in Model view.
- Configured a one-to-many relationship, with Customers on the one side and Orders on the many side.

Step 3: DAX Measures

The following measures summarize the order data. The formulas assume the imported table and column names match those shown in the datasets.

Total Revenue:
```
Total Revenue = SUM('customer_analysis_orders_5000 (1)'[SalesAmount])
```

Total Orders:
```
Total Orders = DISTINCTCOUNT('customer_analysis_orders_5000 (1)'[OrderID])
```

Total Quantity:
```
Total Quantity = SUM('customer_analysis_orders_5000 (1)'[Quantity])
```

Average Order Value:
```
Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)
```

Purchasing Customers:
```
Purchasing Customers = DISTINCTCOUNT('customer_analysis_orders_5000 (1)'[CustomerID])
```

Step 4: Dashboard Development

- Created KPI cards for Total Orders, Total Revenue, Average Order Value, and Purchasing Customers.
- Created a horizontal bar chart for order count by ProductCategory.
- Created a donut chart for order count by PaymentMethod.
- Created a line chart for revenue over the OrderDate timeline.
- Created customer bar charts using Total Revenue and Total Orders to compare customer contribution and purchase frequency.
- Arranged and formatted the visuals on a single report page.

Questions Answered by the Dashboard

- Which product category is most popular? — The ProductCategory bar chart compares order counts by category.
- Which payment method is most frequently used? — The PaymentMethod donut chart compares order counts by payment method.
- How does revenue change month by month? — The revenue trend visual shows revenue over time. For a strictly monthly comparison, group the date axis by Year and Month.
- Which customers have made the most purchases? — The customer order chart compares Total Orders by customer; a Top N filter can display the top ten customers.
- Are a small number of customers responsible for a large proportion of revenue? — The customer revenue chart compares revenue by customer. A Top N revenue-share measure can quantify the contribution of leading customers.

Output

The Power BI report contains KPI cards and charts for product-category orders, payment-method distribution, revenue over time, customer revenue, and customer purchase frequency. The screenshot below documents the dashboard.

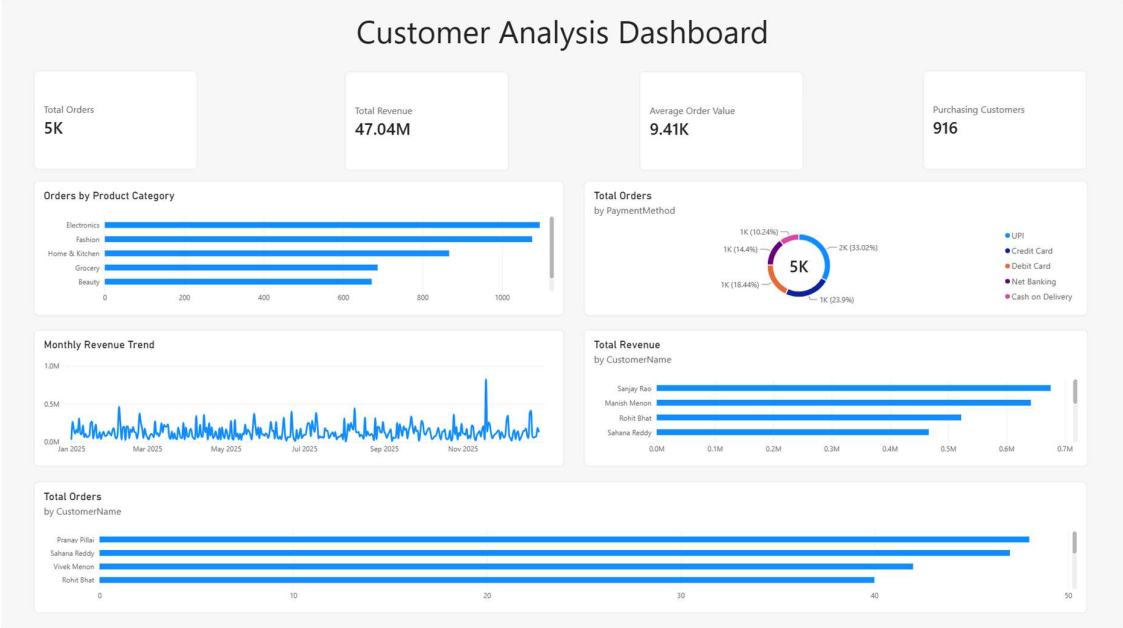


Figure 1: Customer Analysis Dashboard in Power BI

Data Model

The model links the Orders table to the Customers table through CustomerID. This relationship allows customer attributes to be used with order measures in the same report.

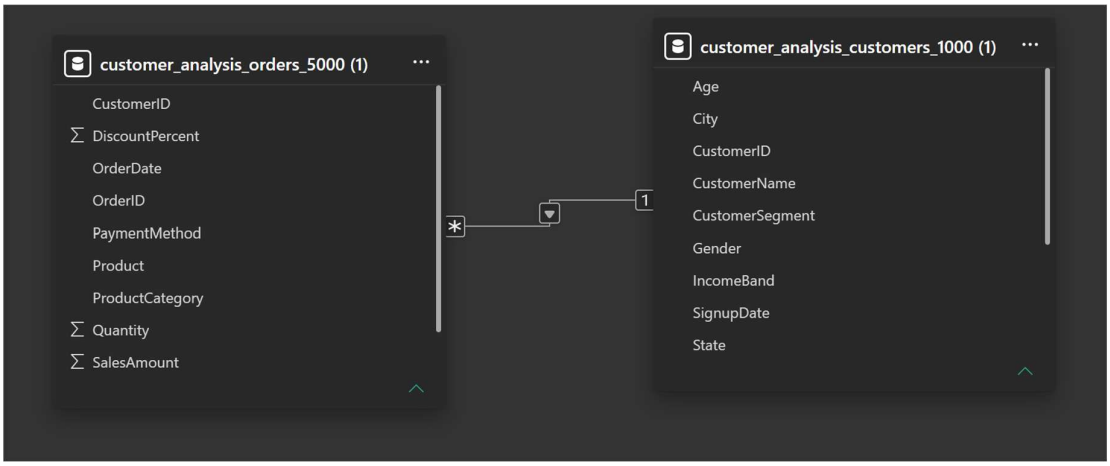


Figure 2: One-to-many relationship between Orders and Customers

Observed Dashboard Summary

- The dashboard displays 5K total orders and 916 purchasing customers.
- The Total Revenue card displays 47.04M, and Average Order Value displays 9.41K using the current card display units.
- The payment-method donut chart shows UPI as the largest share in the displayed visual, at approximately 33.02%.
- The category chart compares order counts by category. Adjust its size or filters to ensure all categories are visible.
- The current revenue line chart shows a date-level pattern. Group the axis by Year and Month before interpreting it as a month-by-month trend.

Limitations

The available order fields shown in the dataset include SalesAmount but do not show a separate cost or profit field. Therefore, actual profit cannot be calculated from the available fields shown; revenue should not be reported as profit. The dashboard should also be checked for complete category and customer visibility, and the date axis should be grouped by month for an accurate monthly trend.

Conclusion

This assignment demonstrates how customer and transaction datasets can be imported into Power BI, connected through CustomerID, and summarized with DAX measures. The dashboard brings together key indicators and visual comparisons for product categories, payment methods, revenue over time, and customer purchasing behaviour. Grouping the date

axis by month and configuring the customer charts for the intended Top N results will make the analysis clearer and more useful for business decision-making.

Learning Outcomes

Through this assignment, I learned:

- How to import CSV datasets into Power BI Desktop.
- How to create and verify a one-to-many relationship between tables.
- How to write DAX measures for revenue, orders, average order value, quantity, and purchasing customers.
- How to select suitable visuals for categorical, time-series, and customer-level analysis.
- How to format and arrange visuals to build a consolidated business dashboard.